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Problems on Double integration.

1] Evaluate $\int_0^1 \int_x^{x^2} xy(x^2+y^2)\,dy\,dx$

→ let $I = \int_0^1 \int_x^{x^2} xy(x^2+y^2)\,dy\,dx$.

$$= \int_0^1 \int_x^{x^2} (x^3y + xy^3)\,dy\,dx.$$

$y=x \quad y=x^2$

$$= \int_0^1 \left[x^3\frac{(x^2)^2}{2} + x\frac{(x^2)^4}{4} \right] - \left(\frac{x^3x^2}{2} + \frac{xx^4}{4} \right)$$

$$= \int_0^1 \left[\frac{x^7}{2} + \frac{x^9}{4} - \frac{3x^5}{4} \right) dx$$

$$= \frac{x^8}{16} + \frac{x^{10}}{40} - \frac{x^6}{8}$$

$$I = \frac{1}{16} + \frac{1}{40} - \frac{1}{8}$$

$$= \frac{5+2-10}{80}$$

$$\boxed{I = \frac{-3}{80}}$$

2] Evaluate $\int_0^1 \int_0^y xye^{x^2}\,dx\,dy$.

→ let $I = \int_0^1 \int_0^y xye^{x^2}\,dx\,dy$

$$= \int_0^1 y \left[\int_0^y xe^{x^2}\,dx \right) dy$$

Put $v = x^2 \quad dv = 2x\,dx \quad = x\,dx = \frac{dv}{2}$

x	0	y
v	0	y

$$\int_0^y x e^{x^2} dx = \frac{1}{2}\int_0^{y^2} e^v dv$$

$$= \frac{1}{2}\left[e^v\right]_0^{y^2}$$

$$= \frac{1}{2}(e^{y^2}-1)$$

$$I = \int_0^1 y\left(\frac{e^{y^2}-1}{2}\right)dy$$

$$= \frac{1}{2}\int_0^1 y(e^{y^2}-1)dy$$

$$= \frac{1}{2}\left(\int_0^1 y e^{y^2} dy - \int_0^1 y\,dy\right)$$

let $t = y^2 \therefore dt = 2y\,dy \Rightarrow y\,dy = \frac{dt}{2}$

limit $0 \to 1$

$$I = \int_0^1 y e^{y^2} dy = \frac{1}{2}\int_0^1 e^t dt$$

$$= \frac{1}{2}\left[e^t\right]_0^1$$

$$= \frac{1}{2}(e-1)$$

$$= \int_0^1 y\,dy = \frac{1}{2}\left[\frac{e-1}{2}\right]_{-2}^{1}$$

$$I = \frac{1}{2}\left(\frac{e-2}{2}\right)$$

$$\boxed{I = \frac{e-2}{4}}$$

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3] Evaluate $\int_0^1 \int_0^x e^{y/x}\, dy\, dx$.

→ let $I = \int_0^1 \int_0^x e^{y/x}\, dy\, dx$.

$$= \int_0^1 \int_{y=0}^{y=x^2} e^{y/x}\, dy\, dx = \int_0^1 \left[\frac{e^{y/x}}{1/x}\right]_0^{x^2} dx = \int_0^1 [x e^{x^2/x} - e^0]\, dx$$

$$= \int_0^1 [x(e^x - 1)]\, dx = \int_0^1 x e^x dx - \int_0^1 x\, dx$$

$$= [xe^x - e^x]_0^1 - \left[\frac{x^2}{2}\right]_0^1$$

$$= [(e-e) - (0-1)] - [(1/2 - 0)]$$

$$= 1 - \frac{1}{2} = \frac{1}{2}$$ //

4] Evaluate $\int_0^1 \int_y^{10y} \sqrt{xy - y^2}\, dx\, dy$

→ let $I = \int_0^1 \int_y^{10y} \sqrt{xy - y^2}\, dx\, dy$.

$xy - y^2 = y(x-y) = \sqrt{y}\sqrt{x-y}$.

$$= \int_0^1 \int_y^{10y} \sqrt{y}\sqrt{x-y}\, dx\, dy$$

let $v = x - y$

$dx = dy$.

x	y	$10y$
v	0	$9y$

$$I = \int_0^{9y} \sqrt{y}\sqrt{v}\, dv$$

$$= \sqrt{y} \int_0^{9y} u^{1/2}\, du$$

$$= \sqrt{y} \left[\frac{2}{3} u^{3/2}\right]_0^{9y}$$

$$= \sqrt{y} \cdot \frac{2}{3} (9y)^{3/2}$$

Ea.

$$= \sqrt{y} \cdot \frac{2}{3} \cdot 27 y^{3/2}$$

$$(9y)^{3/2} = 9^{3/2} y^{3/2} = 27 y^{3/2}$$

$$= 18 y^2$$

$$I = \int_0^1 18 y^2\, dy$$

$$= 18 \left[\frac{y^3}{3}\right]_0^1$$

$$= 18 \times \frac{1}{3}$$

$$\boxed{I = 6}$$

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Problems on Region with given boundaries

1) Evalute $\iint_R y\,dx\,dy$ wher $y = x$ ← line
parabola $= 4x - x^2$

→ region is bounded of the line $y = x$ & parabola $= 4x - x^2$.
The pt. of interaction solving simultaneously

$x = 4x - x^2 \rightarrow x^2 - 3x = 0 \quad x = 0$ & $x = 3$

$y = 0$ & $y = 3$

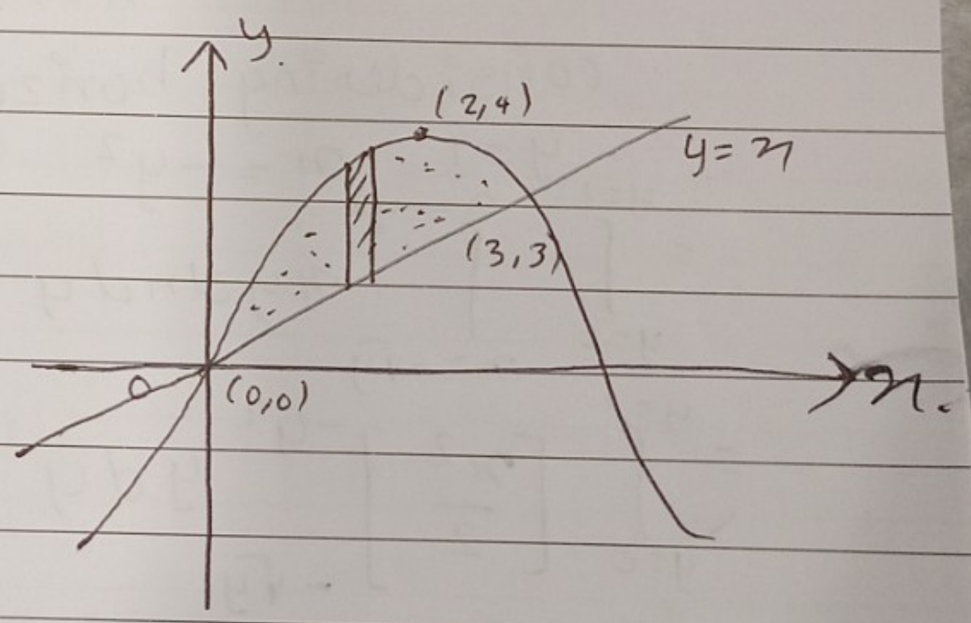
The given equation of the parabola

$$y = 4x - x^2.$$

$y - 4 = -4 + 4x - x^2 \qquad y - 4 = -(x^2 - 4x + 4)$

$y - 4 = -(x-2)^2 \qquad (x-2)^2 = (-y-4)$

$V = (2, 4)$



Taking vertical strip.

$$= \int_{x=0}^{x=3} \int_{y=x}^{y=4x-x^2} y\,dx\,dy.$$

$$= \int_{x=0}^{x=3} \left[\frac{y^2}{2}\right]_x^{4x-x^2} dx$$

$$= \frac{1}{2} \int_{x=0}^{x=3} [(4x - x^2)^2 - x^2]\,dx$$

$$= \frac{1}{2} \int_{x=0}^{x=3} [16x^2 + x^4 - 8x^3 - x^2]\,dx = \frac{1}{2} \int_{x=0}^{x=3} [x^4 + 15x^2 - 8x^3]\,dx.$$

$$= \frac{1}{2}\int_0^3 (x^4 + 15x^2 - 8x^3)\,dx = \frac{1}{2}\left[\left(\frac{x^5}{5}\right)_0^3 + 15\left(\frac{x^3}{3}\right)_0^3 - 8\left(\frac{x^4}{4}\right)_0^3\right]$$

$$= \frac{1}{2}\left[\frac{3^5}{5} + 15\frac{3^3}{3} - 8\frac{3^4}{4}\right] = \frac{1}{2}\left[\frac{243}{5} + (15\times 9) - (2\times 81)\right]$$

$$= \frac{1}{2}\left[\frac{243}{5} + 135 - 162\right] = \frac{1}{2}\left[\frac{243}{5} - 27\right] = \frac{1}{2}\left[\frac{243-135}{5}\right]$$

$$= \frac{1}{2}\left[\frac{108}{5}\right] = \frac{54}{5}.$$

$$\boxed{I = \frac{54}{5}}$$

2] Evaluate $\iint_R xy\,dx\,dy$ region bounded $y = x^2$ parabola $y^2 = -x$

→ The boundary curves of the region R,

$x^2 = y$ $\qquad y^2 = -x$

Pt. of intersection

$(x^2)^2 = -x \Rightarrow x^4 + x = 0 \Rightarrow x(x^3+1) = 0 \quad x = 0, x = -1$

considering horizontal strip.

$y = 1 \quad x = -y^2$

$$= \int_{y=0}^{y=1} \int_{x=-\sqrt{y}}^{x=-y^2} xy\,dx\,dy$$

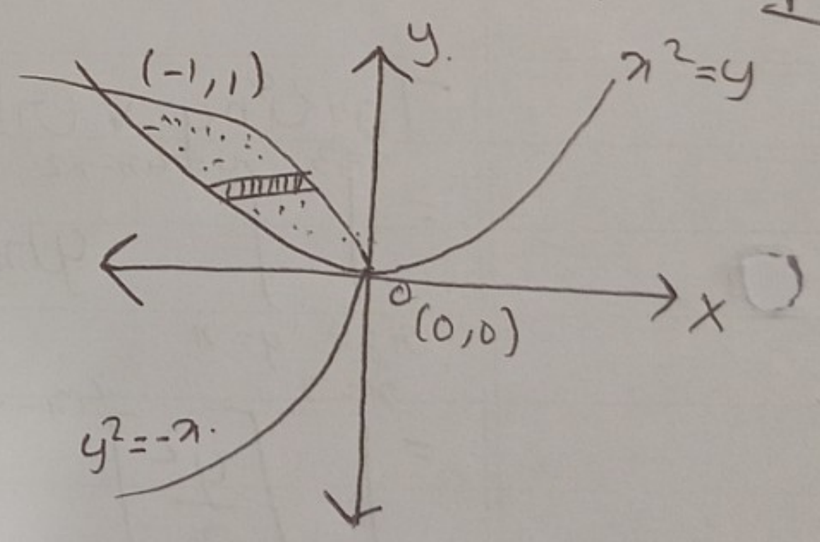
$$= \int_{y=0}^{y=1} \left[\frac{x^2}{2}\right]_{-\sqrt{y}}^{-y^2} y\,dy$$

$$= \int_{y=0}^{y=1} \left[\frac{y^4}{2} - \frac{y}{2}\right] y\,dy$$

$$= \frac{1}{2}\int_{y=0}^{y=1} y^5\,dy - \frac{1}{2}\int_{y=0}^{y=1} y^2\,dy$$

$$= \frac{1}{2}\left(\frac{y^6}{6}\right)_0^1 - \frac{1}{2}\left(\frac{y^3}{3}\right)_0^1 = \frac{1}{2}\left(\frac{1}{6}\right) - \frac{1}{2}\left(\frac{1}{3}\right)$$

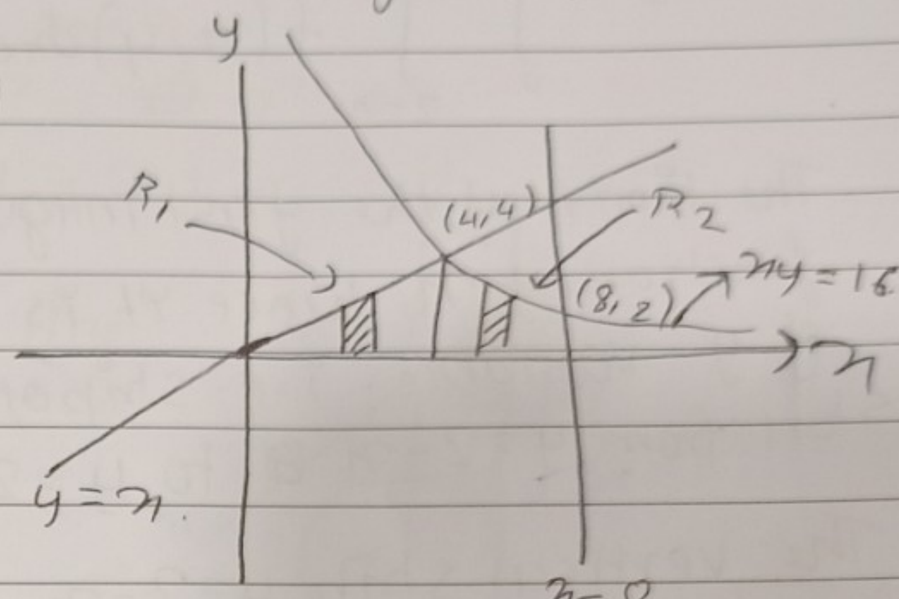
$$= \frac{1}{12} - \frac{1}{6} = \frac{1-2}{12} = -\frac{1}{12}$$



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3] Evaluate $\iint_R x^2\,dx\,dy$ curves $xy=16$, $x=4$, $y=0$ and $x=8$ in first quadrant.

→ $xy=16$ (rectangular hyperbola)
$x=y$ (st. line from origin)
$y=0$ (x-axis)
$x=8$ (line passing to y-axis.



$\therefore y = x$ & $xy = 16$

$x \cdot x = 16 \Rightarrow x^2 = 16 = x \pm 4$

$y = \pm 4$.

$x = 8 \quad xy = 16$

$8y = 16 \Rightarrow y = 2$.

Considering vertical strip.

for $R_1 = \int_{x=0}^{x=4} \int_{y=0}^{y=x} x^2\,dx\,dy$

for $R_2 = \int_{x=4}^{x=8} \int_{y=0}^{y=16/x} x^2\,dx\,dy$

$$\therefore \int_{x=0}^{x=4} \int_{y=0}^{y=x} x^2\,dx\,dy + \int_{x=4}^{x=8} \int_{y=0}^{y=16/x} x^2\,dx\,dy = \int_{x=0}^{x=4} x^2 (y)_0^x\,dx + \int_{x=4}^{x=8} x^2 (y)_0^{16/x}\,dx$$

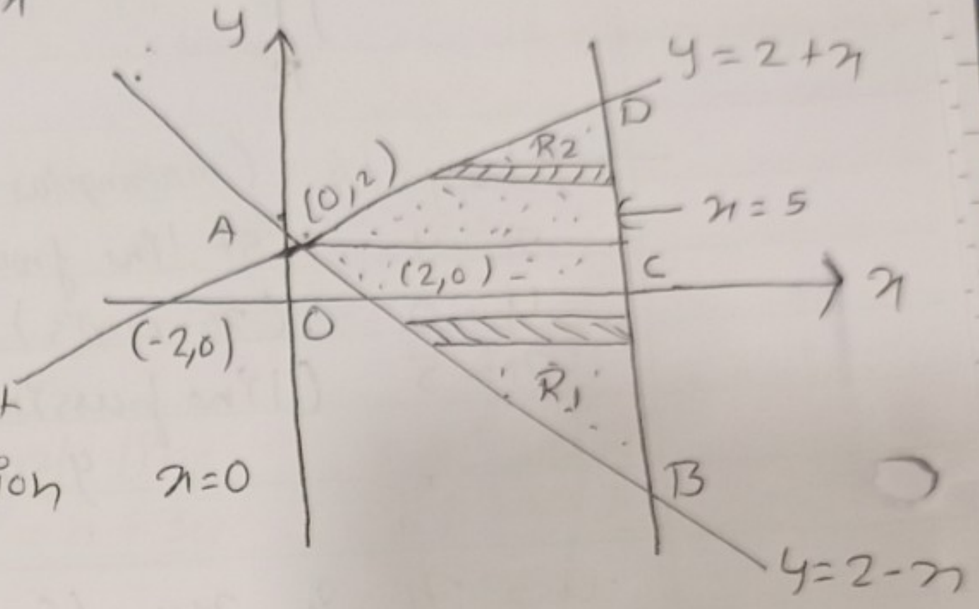
$$= \int_{x=0}^{x=4} x^3\,dx + \int_{x=4}^{x=8} x^2 \frac{16}{x}\,dx = \left(\frac{x^4}{4}\right)_0^4 + 16\left(\frac{x^2}{2}\right)_4^8$$

$64 + \frac{16}{2}(64-16) = 64 + 384 = \underline{\underline{448}}$

Problems on change of order.

1) change the order of $\int_0^5 \int_{2-x}^{2+x} f(x,y)\,dx\,dy$

$\longrightarrow$ let, $I = \int_0^5 \int_{2-x}^{2+x} f(x,y)\,dx\,dy$



The limit of the first integral is the function of x hence it is the limit of y-variable. i.e. strip of the region start from $y = 2-x$ to $y = 2+x$

The vertical strip $y = 2-x$ to $y = 2+x$ is moved from $x = 0$ to $x = 5$ to cover the region.

∴ The region of integration is bounded by four curve.

$x = 0$, $x = 5$, $y = 2-x$ & $y = 2+x$.

co-ordinate of B of $x = 5$, $y = 2-x$

$y = 2-5 = -3$ $(5,-3)$

co-ordinate of C is $(5,2)$

& D, $x = 5$

$y = 2+x$ ∴ $y = 2+5 = 7$

co-ordinate of $D(5,7)$

for $R_1 = \int_{y=-3}^{y=2} \int_{x=2-y}^{x=5} f(x,y)\,dx\,dy$ for $R_2 = \int_{y=2}^{y=7} \int_{x=y-2}^{x=5} f(x,y)\,dx\,dy$

∴ On changing the order of integration, we get.

$= \int_{y=-3}^{y=2} \int_{x=2-y}^{x=5} f(x,y)\,dx\,dy + \int_{y=2}^{y=7} \int_{x=y-2}^{x=5} f(x,y)\,dx\,dy$ //.

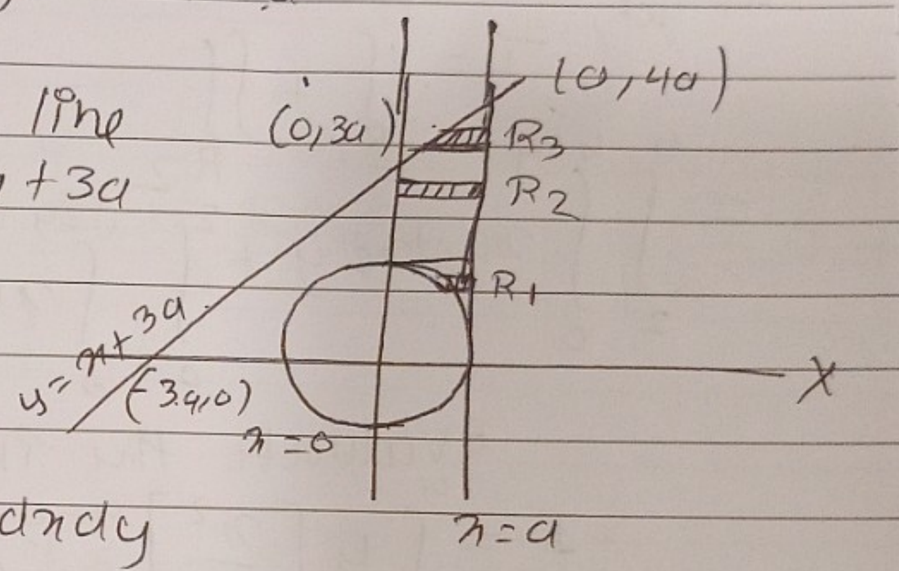
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2) Change the order of integration $\int_0^a \int_{\sqrt{a^2-x^2}}^{x+3a} f(x,y)\,dy\,dx$.

$\rightarrow y = \sqrt{a^2-x^2}$ & $y = x+3a$

$x^2+y^2 = a^2$ $\quad y^2 = a^2 - x^2$

$x=0$ to $x=0$ $\quad y = x+3a$



The pt. of intersection of line
$y = x+3a$ & $x=a \Rightarrow y = a+3a$
$y = 4a$

Taking horizontal strip.

for $R_1 = \int_{y=0}^{y=a} \int_{x=\sqrt{a^2-y^2}}^{x=a} f(x,y)\,dx\,dy$

for $R_2 \int_{y=a}^{y=3a} \int_{x=0}^{x=a} f(x,y)\,dx\,dy$

for $R_3 = \int_{y=3a}^{y=4a} \int_{x=y-3a}^{x=a} f(x,y)\,dx\,dy$

On changing the order of integration we get,

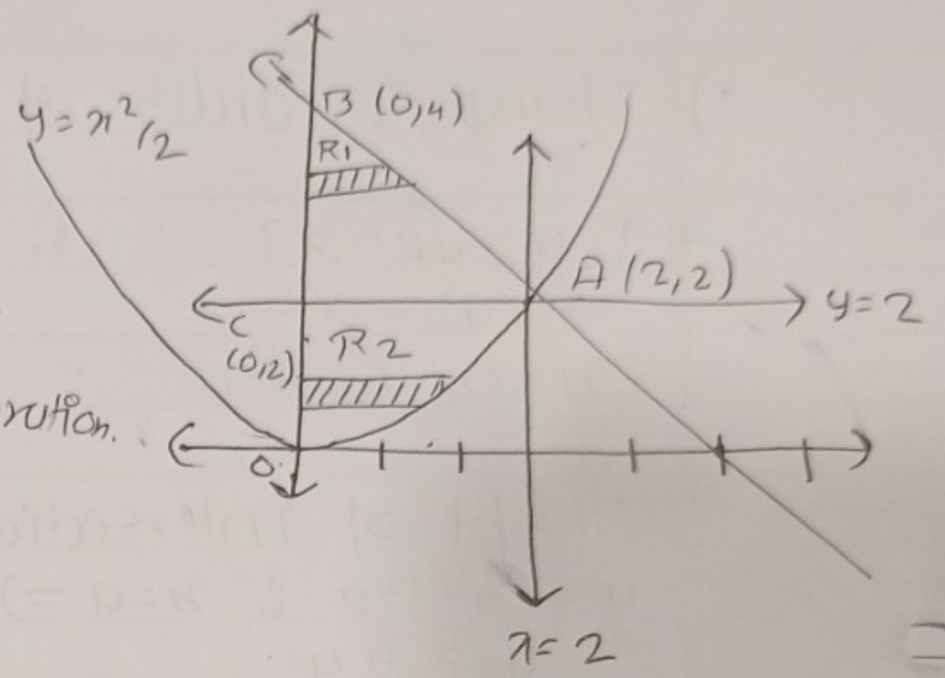
$= \int_{y=0}^{y=a} \int_{x=\sqrt{a^2-y^2}}^{x=a} f(x,y)\,dx\,dy + \int_{y=a}^{y=3a} \int_{x=0}^{x=a} f(x,y)\,dx\,dy$

$+ \int_{y=3a}^{y=4a} \int_{x=y-3a}^{x=a} f(x,y)\,dx\,dy$.

3) change the order of integration $\int_0^2 \int_{x^2/2}^{4-x} xy\,dx\,dy$

→ let $I = \int_0^2 \int_{x^2/2}^{4-x} xy\,dx\,dy$

for R_1 & R_2



for R_1 $x: 0$ to $4-y$ & $y: 2$ to 4

R_2 $x: 0$ to $\sqrt{2y}$ & $y: 0$ to 2

By changing the order of integration. we get,

$$I = \iint_{R_1} + \iint_{R_2}$$

$$= \int_2^4 \int_0^{4-y} xy\,dx\,dy + \int_0^2 \int_0^{\sqrt{2y}} xy\,dx\,dy$$

evaluate the integrals.

$$I = \int_2^4 y \left[\frac{x^2}{2}\right]_0^{4-y} dy + \int_0^2 y \left[\frac{x^2}{2}\right]_0^{\sqrt{2y}} dy$$

$$= \int_2^4 y \left[\frac{(4-y)^2}{2}\right] dy + \int_0^2 y \left[\frac{2y}{2}\right] dy$$

$$= \int_2^4 \frac{y}{2}(16 - 8y + y^2)\,dy + \int_0^2 y^2\,dy$$

$$= \frac{1}{2}\int_2^4 (16y - 8y^2 + y^3)\,dy + \int_0^2 y^2\,dy$$

$$= \frac{1}{2}\left[\frac{16y^2}{2} - \frac{8y^3}{3} + \frac{y^4}{4}\right]_2^4 + \left[\frac{y^3}{3}\right]_0^2 dy$$

$$= \frac{1}{2}\left[8(4)^2 - \frac{8}{3}(4)^3 + \frac{1}{4}(4)^4\right] + \left[\frac{8}{3}\right] - \frac{1}{2}\left[8(2)^2 - 8(2)^3 + \frac{1}{4}(2)^4\right]$$

$$= \frac{1}{2}\left[128 - \frac{512}{3} + 64 - 32 - \frac{64}{3} + 4\right] + \frac{8}{3}$$

$$I = \frac{1}{2}\left[\frac{20}{3}\right] + \frac{8}{3} = \underline{6}$$

$\boxed{I = 6}$

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Problems on Polar co-ordinate system.

1) Evaluate $\iint_R x^2y^2\,dx\,dy$ where R is the region enclosed by ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$

$\longleftrightarrow$

$x = ar\cos\theta \qquad y = br\sin\theta$

$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1 \Rightarrow \frac{a^2r^2\cos^2\theta}{a^2}+\frac{b^2r^2\sin^2\theta}{b^2}=1$

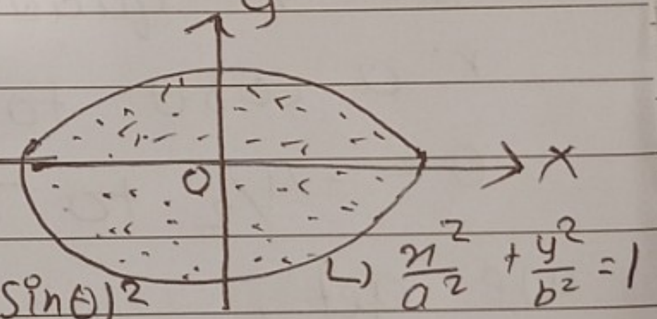
$r^2[\cos^2\theta+\sin^2\theta]=1$

$r^2=1 \Rightarrow r=1$

$$\begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} a\cos\theta & -ar\sin\theta \\ b\sin\theta & br\cos\theta \end{vmatrix}$$

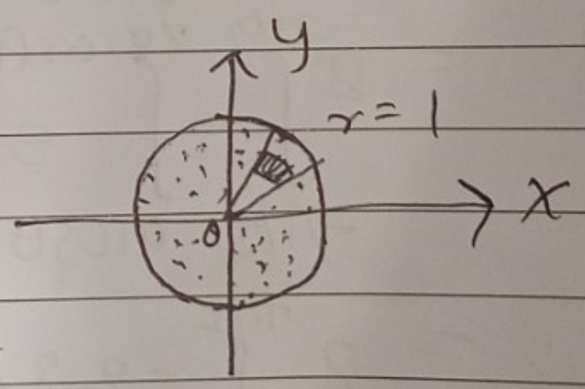
$= abr\cos^2\theta + abr\sin^2\theta$

$= abr(\cos^2\theta+\sin^2\theta) = abr$



$\therefore \iint x^2y^2\,dx\,dy = \iint (ar\cos\theta)^2(br\sin\theta)^2\,(abr)\,dr\,d\theta$

This integration is to be evaluated on circle $r=1$



$= a^3b^3\int_{\theta=0}^{2\pi}\int_{r=0}^{1}\cos^2\theta\sin^2\theta\,r^5\,dr\,d\theta$

$= a^3b^3\int_0^{2\pi}\left[\frac{r^6}{6}\right]_0^1\cos^2\theta\sin^2\theta\,d\theta$

$= \frac{a^3b^3}{6}\int_0^{2\pi}\cos^2\theta\cdot\sin^2\theta\,d\theta = \frac{a^3b^3}{6}\,2\int_0^{\pi}\cos^2\theta\sin^2\theta\,d\theta$

$$= \frac{a^3b^3}{6} \cdot 4 \int_0^{\pi/2} \cos^2\theta \cdot \sin^2\theta \, d\theta = \frac{2a^3b^3}{3} \frac{1}{2} \beta\left(\frac{3}{2}, \frac{3}{2}\right) \quad \cdots \left[\because \int_0^{2a} f(x)dx = 2\int_0^a f(x)dx \text{ if } f(2a-x) = f(x)\right]$$

$$= \frac{a^3b^3}{3} \frac{\sqrt{\frac{3}{2}}\sqrt{\frac{3}{2}}}{\sqrt{3}} = \frac{a^3b^3}{3} \frac{\frac{1}{2}\sqrt{\frac{1}{2}}\frac{1}{2}\sqrt{\frac{1}{2}}}{2} = \frac{a^3b^3\pi}{24} //.$$

2) Evaluate $\iint_R y^2 dx\,dy$ over the area $x^2+y^2-ax=0$ and in R

$\rightarrow$

$$I = \iint_R y^2 dx\,dy.$$

R is the region bounded by $x^2+y^2-ax=0$ and $x^2+y^2-2ax=0$

By changing the co-ordinates,

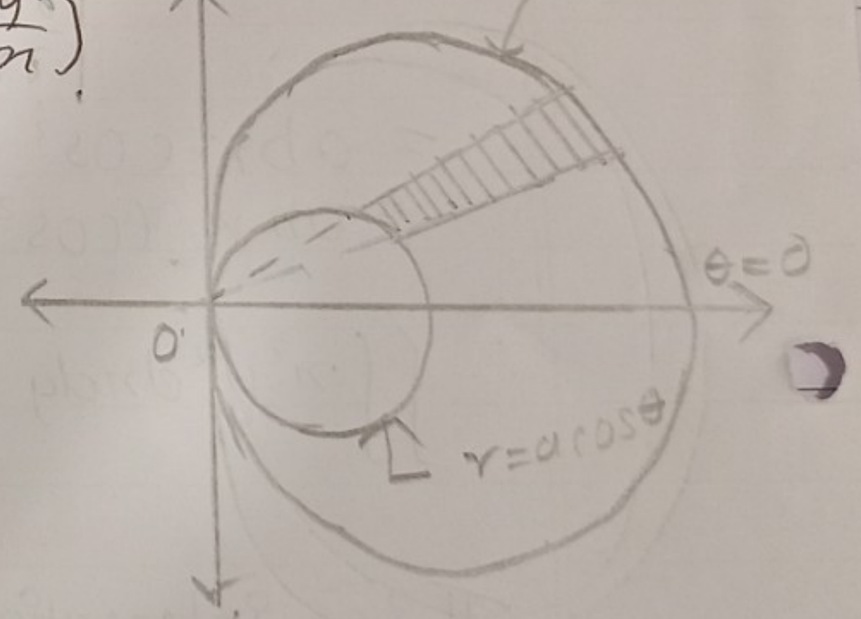
$x = r\cos\theta$, $y = r\sin\theta$, $dx\,dy = r\,dr\,d\theta$

where $r^2 = x^2+y^2$ & $\theta = \tan^{-1}\left(\frac{y}{x}\right)$

for 1st quadrant

$r: a\cos\theta$ to $2a\cos\theta$

$\theta = -\pi/2$ to $\pi/2$



we get,

$$I = 2\int_{-\pi/2}^{\pi/2} \int_{a\cos\theta}^{2a\cos\theta} (r\sin\theta)^2 r\,dr\,d\theta$$

$$= 2\int_{-\pi/2}^{\pi/2} \int_{a\cos\theta}^{2a\cos\theta} \sin^2\theta\, r^3\, dr\, d\theta$$

$$= 2\int_{-\pi/2}^{\pi/2} \sin^2\theta \left[\frac{r^4}{4}\right]_{a\cos\theta}^{2a\cos\theta} d\theta =$$

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$$= \frac{1}{2}\int_{-\pi/2}^{\pi/2} \sin^2\theta\left[16a^4\cos^4\theta - a^4\cos^4\theta\right] d\theta$$

$$= \frac{15a^4}{2}\int_{-\pi/2}^{\pi/2} \sin^2\theta \cdot \cos^4\theta \, d\theta = \frac{15a^4}{2}\,\frac{1}{2}\,\beta\left(\frac{3}{2},\frac{5}{2}\right)$$

$$= \frac{15a^4}{4}\,\frac{\Gamma 3/2\,\Gamma 5/2}{\Gamma 4} = \frac{15\pi a^4}{64}$$

$$\boxed{I = \frac{15\pi a^4}{64}}$$

3) Evaluate $\iint_R \frac{dx\,dy}{(1+x^2+y^2)^2}$ where R is the region bounded by one loop $(x^2+y^2)^2 = x^2 - y^2$

$\rightarrow I = \iint_R \frac{dx\,dy}{(1+x^2+y^2)^2}$ where R is the region bounded by one loop.

$(x^2+y^2)^2 = x^2 - y^2$

$x = r\cos\theta$ & $y = r\sin\theta$, we get

$(r^2)^2 = r^2(\cos^2\theta - \sin^2\theta)$

$r^4 = r^2\cos 2\theta$ $\quad\therefore r^2 = \cos 2\theta$

$r = \sqrt{\cos 2\theta}$ $\quad\therefore dx\,dy = r\,dr\,d\theta$.

where $r = \sqrt{x^2+y^2}$ & $\theta = \tan^{-1}(y/x)$

we consider

$\theta = -\pi/4$ to $\pi/4$ $\quad r = 0$ to $\sqrt{\cos 2\theta}$

we have

$$I = 2\int_{-\pi/4}^{\pi/4}\int_{0}^{\sqrt{\cos 2\theta}} \frac{r\,dr\,d\theta}{(1+r^2)^2}$$

$$I = 2\int_{-\pi/4}^{\pi/4}\int_{0}^{\sqrt{\cos 2\theta}} \frac{r\,dr\,d\theta}{(1+r^2)^2} = \int_{-\pi/4}^{\pi/4}\int_{0}^{\sqrt{\cos 2\theta}} \frac{2r\,dr}{(1+r^2)^2}\,d\theta$$

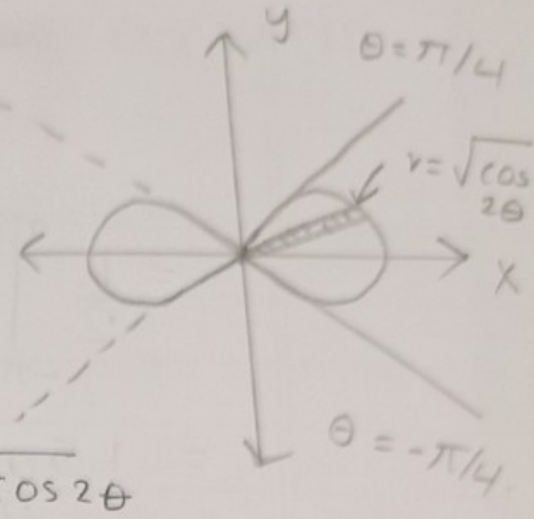
$$= \int_{-\pi/4}^{\pi/4}\int_{0}^{\sqrt{\cos 2\theta}} \frac{r\,dr\,d\theta}{(1+r^2)^2} = \int_{-\pi/4}^{\pi/4}\left[\frac{-1}{1+r^2}\right]_0^{\sqrt{\cos 2\theta}} d\theta$$

$$= \int_{-\pi/4}^{\pi/4}\left[1 - \frac{1}{1+\cos 2\theta}\right] d\theta$$

$$= \int_{-\pi/4}^{\pi/4}\left[1 - \frac{1}{2\cos^2\theta}\right] d\theta = \int_{-\pi/4}^{\pi/4}\left[1 - \frac{1}{2}\sec^2\theta\right] d\theta$$

$$I = \left[\theta - \frac{1}{2}\tan\theta\right]_0^{\pi/4} = \frac{\pi}{4} - \frac{1}{2}\tan\left(\frac{\pi}{4}\right) = \frac{\pi}{4} - \frac{1}{2}$$

$$\boxed{I = \frac{\pi}{4} - \frac{1}{2}}$$

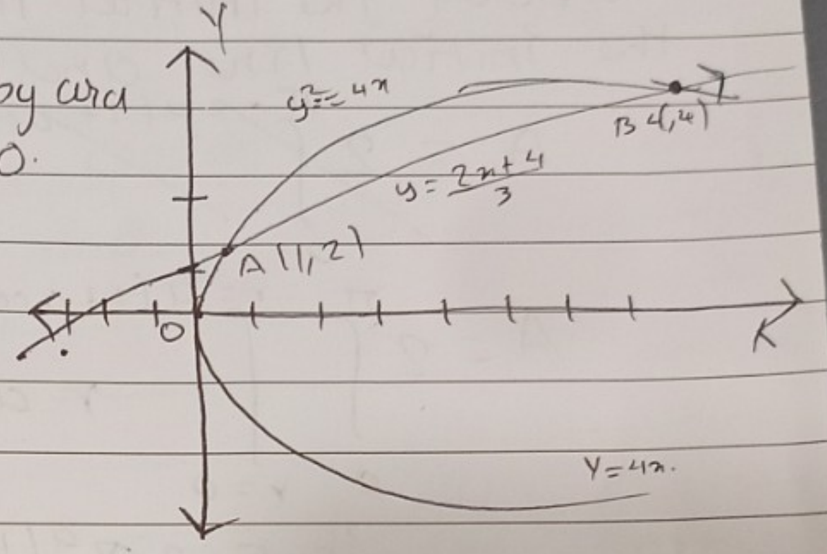


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Problems on area using double integration.

1) Find the area bounded by $y^2 = 4x$ & $2x - 3y + 4 = 0$

→



let A be the area bounded by curve $y^2 = 4x$ & line $2x - 3y + 4 = 0$.

$\frac{y^2}{2} = 2x$, consider.

$\left(\frac{y^2}{2}\right) - 3y + 4 = 0$

$y^2 - 6y + 8 = 0$

$(y-2)(y-4) = 0$

$y = 2$ or $y = 4$

if $y = 2$ then $4 = 4x$ $\therefore x = 1$

if $y = 4$ then $16 = 4x$ $\therefore x = 4$

The point of intersection are $(1,2)$ & $(4,4)$

x: 1 to 4

y: $\frac{2x+4}{3}$ to $2\sqrt{x}$.

we get,

$$A = \int_1^4 \int_{\frac{2x+4}{3}}^{2\sqrt{x}} dy\,dx = \int_1^4 [y]_{\frac{2x+4}{3}}^{2\sqrt{x}}\,dx.$$

$$A = \int_1^4 \left(2\sqrt{x} - \frac{2x+4}{3}\right)dx = \int_1^4 \left[2\sqrt{x} - \frac{2}{3}x - \frac{4}{3}\right]dx$$

$$= \left[2\frac{x^{3/2}}{3/2} - \frac{2}{3}\frac{x^2}{2} - \frac{4}{3}x\right]_1^4 = \left[\frac{4}{3}x^{3/2} - \frac{1}{3}x^2 - \frac{4}{3}x\right]_1^4$$

$$= \left(\frac{4}{3}\cdot 4 - \frac{16}{3} - \frac{16}{3}\right) - \left(\frac{-1}{3}\right) = \frac{1}{3}$$

$$A = \frac{1}{3} \text{ sq. unit.}$$

2) Find the area of cardiode $r = a(1+a\cos\theta)$

→ since cardiode $r = a(1+\cos\theta)$ is symmetrical about the initial line. so we can find the area above the initial line and double it to get required area.

$$A = 2\int_0^{\pi} \int_{r=0}^{r=a(1+a\cos\theta)} r\,dr\,d\theta$$

$$= 2\int_0^{\pi} \left[\frac{r^2}{2}\right]_0^{a(1+\cos\theta)} d\theta.$$

$$= \int_0^{\pi} a^2(1+\cos\theta)^2 d\theta$$

$$= a^2 \int_0^{\pi} (1+\cos^2\theta + 2\cos\theta)\,d\theta$$

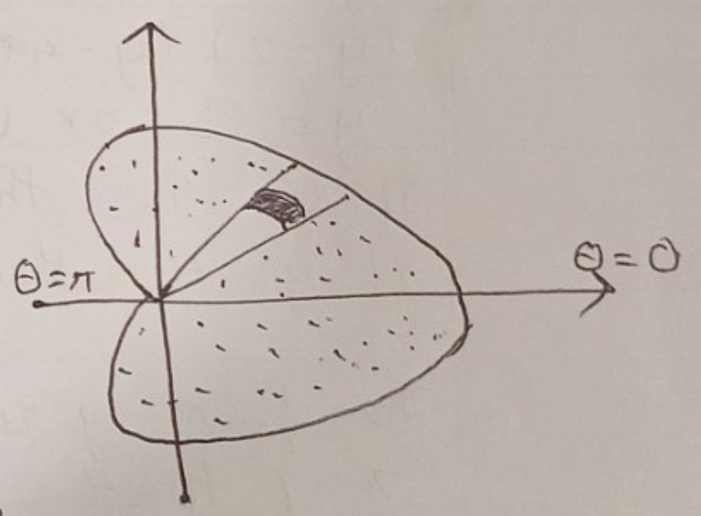
$$= a^2\left[\int_0^{\pi} d\theta + \int_0^{\pi} \cos^2\theta\,d\theta + 2\int_0^{\pi} \cos\theta\,d\theta\right]$$

$$= a^2\left[(\theta)_0^{\pi} + 2\int_0^{\pi/2} \cos^2\theta\,d\theta + 2(\sin\theta)_0^{\pi}\right]$$

$$= a^2\left[\pi + 2\cdot\frac{1}{2}\beta\left(\frac{3}{2},\frac{1}{2}\right) + 2(0-0)\right]$$

$$= a^2\left[\pi + \frac{\sqrt{\frac{3}{2}}\sqrt{\frac{1}{2}}}{\sqrt{2}}\right]$$

$$= a^2\left[\pi + \frac{\frac{1}{2}\sqrt{\frac{1}{2}}\sqrt{\frac{1}{2}}}{1!}\right] = a^2\left[\pi + \frac{\pi}{2}\right] = \frac{3\pi a^2}{2} //$$



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3) find the area common to the circles $r = 2\cos\theta$ and $r = 2\sin\theta$.

→

let A be the area common to the circles $r = 2\cos\theta$ & $r = 2\sin\theta$.

for point of intersection, we consider

$2\cos\theta = 2\sin\theta$, $\cos\theta = \sin\theta$

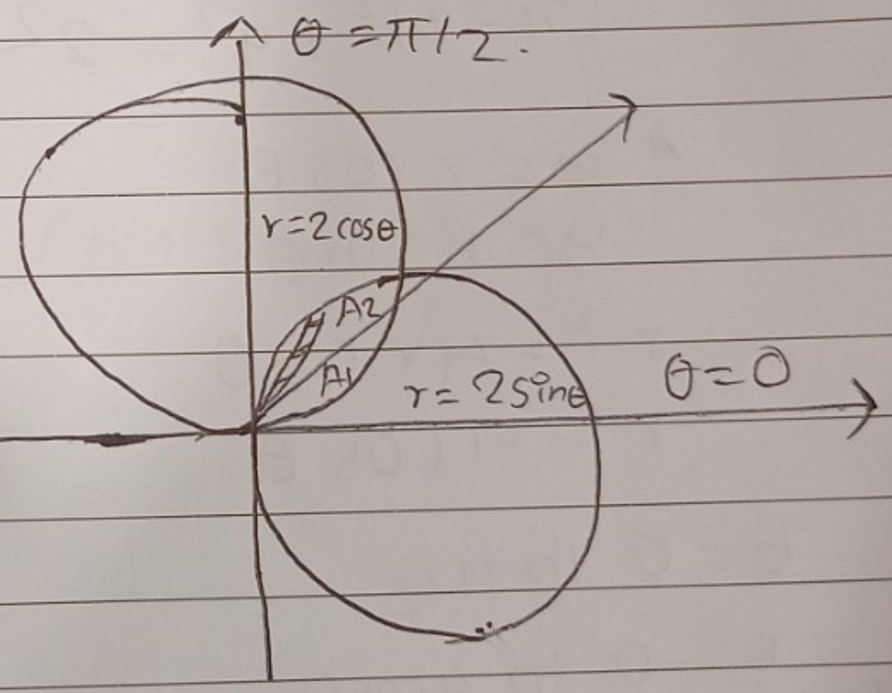
$1 = \tan\theta$ $\therefore \theta = \tan^{-1}1 = \pi/4$

$\theta = \frac{\pi}{4}$

Consider the Area A_1, for the circle $r = 2\sin\theta$ to $\theta = \frac{\pi}{4}$ and area A_2 for the circle.

~~Consider the Area A2~~

$r = 2\cos\theta$ from $\theta = \frac{\pi}{4}$ to $\frac{\pi}{2}$



for Area A_1 $\theta = 0$ to $\frac{\pi}{4}$

for Area A_2 $r: 0$ to $2\sin\theta$.

we have $A = A_1 + A_2$.

$$A = \int_0^{\pi/4}\int_0^{2\sin\theta} r\,dr\,d\theta + \int_{\pi/4}^{\pi/2}\int_0^{2\cos\theta} r\,dr\,d\theta$$

$$= \int_0^{\pi/4}\left[\frac{r^2}{2}\right]_0^{2\sin\theta} d\theta + \int_{\pi/4}^{\pi/2}\left[\frac{r^2}{2}\right]_0^{2\cos\theta} d\theta$$

$$= \int_0^{\pi/2}[2\sin^2\theta]\,d\theta + \int_{\pi/4}^{\pi/2}(2\cos^2\theta)\,d\theta = \int_0^{\pi/4}(1-\cos 2\theta)\,d\theta +$$

$$\int_{\pi/4}^{\pi/2} (1+\cos 2\theta)\,d\theta$$

$$= \left[\theta - \frac{\sin 2\theta}{2}\right]_0^{\pi/4} + \left[\theta + \frac{\sin 2\theta}{2}\right]_{\pi/4}^{\pi/2}$$

$$= \frac{\pi}{4} - \frac{1}{2}\sin\left(\frac{\pi}{2}\right) + \frac{\pi}{2} + 0 - \frac{\pi}{4} - \frac{1}{2}\sin\left(\frac{\pi}{2}\right) = \frac{\pi}{2} - 1$$

$A = \frac{\pi}{2} - 1$ sq units

4) Find the area bounded by the circle $x^2+y^2=4x$

→

let $A = x^2+y^2=4x$

$x^2-4x+4+y^2=4$

$(x-2)^2+y^2=2^2$

$x = r\cos\theta$

$y = r\sin\theta$

$r^2 = 4r\cos\theta$

$r = 4\cos\theta$

$\theta = 0$ to $\pi/2$

$r = 0$ to $4\cos\theta$

$$A = 2\int_0^{\pi/2}\int_0^{4\cos\theta} r\,dr\,d\theta$$

$$= 2\int_0^{\pi/2}\left[\frac{r^2}{2}\right]_0^{4\cos\theta} d\theta$$

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$$= 2\int_0^{\pi/2} \frac{16\cos^2\theta}{2}\,d\theta$$

$$= 16\int_0^{\pi/2} \cos^2\theta\,d\theta$$

$$= 16\int_0^{\pi/2} \frac{1+\cos 2\theta}{2}\,d\theta$$

$$= 8\int_0^{\pi/2} (1+\cos(2\theta)\,d\theta$$

$$= 8\left[\theta + \frac{\sin(2\theta)}{2}\right]_0^{\pi/2}$$

$$= 8\left(\left(\frac{\pi}{2} + \frac{\sin(\pi)}{2}\right) - \left(0 + \frac{\sin(0)}{2}\right)\right)$$

$$= 8\left(\frac{\pi}{2} + 0 - 0\right) = 4\pi$$

$$A = \underline{\underline{4\pi}}$$

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Problems on Triple Integration

1) Evaluate $\int_0^{\log 2} \int_0^{x} \int_0^{x+y} e^{x+y+z}\, dx\, dy\, dz$

$\longrightarrow = \int_0^{\log 2} \int_0^{x} \int_{z=0}^{z=x+y} e^x e^y e^z\, dx\, dy\, dz$ 1st integral

$= \int_0^{\log 2} \int_0^{x} [e^z]_0^{x+y} e^x e^y\, dx\, dy$

$= \int_0^{\log 2} \int_0^{x} [e^{x+y} - 1] e^x e^y\, dx\, dy = \int_0^{\log 2} \int_0^{x} [e^x e^y - 1] e^x e^y\, dx\, dy$

$= \int_0^{\log 2} \int_0^{x} [e^{2x} e^{2y} - e^x e^y]\, dx\, dy.$

$\log 2$ 2nd Integral

$= \int_0^{\log 2} \left[e^{2x} \left(\frac{e^{2y}}{2}\right)_0^x - e^x (e^y)_0^x\right] dx = \int_0^{\log 2} \left[e^{2x} \left(\frac{e^{2x} - 1}{2}\right) - e^x (e^x - 1)\right] dx$

$= \int_0^{\log 2} \left[\frac{1}{2} e^{4x} - \frac{1}{2} e^{2x} - e^{2x} + e^x\right] dx = \int_0^{\log 2} \left[\frac{1}{2} e^{4x} - \frac{3}{2} e^{2x} + e^x\right] dx$

$= \frac{1}{2} \left[\frac{e^{4x}}{4}\right]_0^{\log 2} - \frac{3}{2} \left[\frac{e^{2x}}{2}\right]_0^{\log 2} + [e^x]_0^{\log 2}.$

$= \frac{1}{8} [e^{4 \log 2} - e^0] - \frac{3}{4} [e^{2 \log 2} - e^0] + [e^{\log 2} - 1]$

$= \frac{1}{8} [e^{\log 2^4} - 1] - \frac{3}{4} [e^{\log 2^2} - 1] + [2 - 1]$

$= \frac{1}{8} [2^4 - 1] - \frac{3}{4} [2^2 - 1] + [2 - 1]$

$= \frac{15}{8} - \frac{9}{4} + 1 = \frac{15 - 18 + 8}{8} = 2\frac{3-18}{8} = \frac{5}{8}$ //.

2] Evaluate $\int_0^1 \int_{y^2}^1 \int_0^{1-x} x\,dx\,dy\,dz$

$\rightarrow I = \int_0^1 \int_{y^2}^1 \int_0^{1-x} x\,dx\,dy\,dz$.

integrate it first w.r.t z from $z=0$ to $z=1-x$

$= \int_0^1 \int_{y^2}^1 x\,[z]_0^{1-x}\,dx\,dy = \int_0^1 \int_{y^2}^1 x(1-x)\,dx\,dy$.

integrate second integral w.r.t. to x treating y as constant

$= \int_0^1 \int_{y^2}^1 (x - x^2)\,dx\,dy = \int_0^1 \left[\frac{x^2}{2} - \frac{x^3}{3}\right]_{y^2}^1 dy$

$= \int_0^1 \left[\left(\frac{1}{2} - \frac{1}{3}\right) - \left(\frac{y^4}{2} - \frac{y^6}{3}\right)\right] dy = \int_0^1 \left[\frac{1}{6} - \frac{y^4}{2} + \frac{y^6}{3}\right] dy$

$= \frac{1}{6}(y)_0^1 - \frac{1}{2}\left(\frac{y^5}{5}\right)_0^1 + \frac{1}{3}\left(\frac{y^7}{7}\right)_0^1 = \frac{1}{6} - \frac{1}{10} + \frac{1}{21}$

$\boxed{I = \frac{4}{35}}$

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3) Evaluate $\int_0^1 \int_0^{1-x} \int_0^{1-x-y} \frac{dz\,dy\,dx}{(1+x+y+z)^3}$

$\rightarrow$ let $I = \int_0^1 \int_0^{1-x} \int_0^{1-x-y} \frac{dz\,dy\,dx}{(1+x+y+z)^3}$

integrate first wrt to z treat x & y as constant

$= \int_0^1 \int_0^{1-x} \left[\frac{(1+x+y+z)^{-2}}{(-2)}\right]_0^{1-x-y} dy\,dx$

$= \int_0^1 \int_0^{1-x} \left[\frac{(1+x+y+1-x-y)^{-2} - (1+x+y)^{-2}}{(-2)}\right] dy\,dx$

$= \int_0^1 \int_0^{1-x} \left[\frac{(2)^{-2} - (1+x+y)^{-2}}{(-2)}\right] dy\,dx = \int_0^1 \int_0^{1-x} \left[\frac{1}{2(1+x+y)^2} - \frac{1}{8}\right] dy\,dx.$

Integrate wrt to y treating x as constant.

$= \frac{1}{2} \int_0^1 \int_0^{1-x} (1+x+y)^{-2}\,dy\,dx - \frac{1}{8} \int_0^1 \int_0^{1-x} dy\,dx$

$= \frac{1}{2} \int_0^1 \left[\frac{(1+x+y)^{-1}}{-1}\right]_0^{1-x} dx - \frac{1}{8} \int_0^1 (y)_0^{1-x} dx$

$= \frac{1}{2} \int_0^1 \left[\frac{(1+x+1-x)^{-1} - (1+x)^{-1}}{-1}\right] dx - \frac{1}{8} \int_0^1 (1-x)\,dx$

$= \frac{1}{2} \int_0^1 \left[\frac{1}{(1+x)} - \frac{1}{2}\right] dx - \frac{1}{8} \left[x - \frac{x^2}{2}\right]_0^1$

$$= \frac{1}{2} (\log(1+x))_0^1 (x)_0^1 - \frac{1}{8}\left(1 - \frac{1}{2}\right)$$

$$= \frac{1}{2}(\log 2 - 0) - \frac{1}{4} - \frac{1}{8}\,\frac{1}{2}$$

$$\boxed{I = \frac{1}{2}\log 2 - \frac{5}{16}}$$